

Pharm LMS — Project Report

Product: Pharm LMS (PharmLMS)
Type: Web-based Learning Management System for pharmacy and clinical training
Stack: Next.js 16 · React 19 · PostgreSQL · Prisma · NextAuth · Paystack · Cloudflare R2
Report date: May 2026

1. Executive summary

Pharm LMS is a company-owned learning platform that connects **students, tutors (course authors), mentors (1:1 coaching),** and **administrators** in a single product. Learners discover courses, pay where required, progress through structured curriculum, complete assessments, book meetings, and earn badges and **digital certificates**. Tutors build and submit courses for approval; admins govern quality, users, payments, and platform settings.

The platform is production-oriented: role-based portals, Paystack checkout and tutor payouts, media on object storage, and a modern maintainable codebase suitable for ongoing feature work and deployment on Vercel.

2. Platform overview

Portal	Route prefix	Primary users
Student	/student	Learners
Tutor	/tutor	Course creators
Mentor	/mentor	1:1 coaches
Admin	/admin	Operations & governance

Role model: Tutors own courses (`Course.mentorId`). Mentors handle meetings and profile approval only. Students consume content. Admins approve courses, manage CRM, payments, badges, and messaging.

3. Student features (implemented)

3.1 Learning journey

- **Catalog & search** — Browse published courses, filters, wishlist.
- **Enrollment** — Free enroll or **Paystack** paid checkout (`react-paystack` inline popup).
- **Course player** — Video and article lessons, sidebar curriculum, lesson progress, prev/next navigation.
- **Section quizzes** — Modal quizzes with scoring and attempt history.
- **AI quiz** — Generated practice from course content (Hugging Face, configurable).
- **AI course assistant** — Context-aware chat after completing sections.
- **Assignments** — Cross-course hub: submit text and files; view grades and feedback.
- **Forums & announcements** — Course discussion and tutor announcements (gated by enrollment and payment).
- **Reviews** — Star ratings and comments when eligible.
- **Lesson notes** — Per-lesson student notes.
- **Course completion** — Manual “Complete course” when curriculum is finished; celebration modal with tutor congratulatory content (video/article).

3.2 Recognition & credentials

- **Badges** — Rule-based achievements (enrollment, lessons, quizzes, streaks, meetings); toast on earn; dedicated achievements page.
- **Leaderboard** — Rankings from lessons, quiz scores, badges, and streaks.
- **Digital certificates (new)** — On course completion, the system issues a **Pharm LMS certificate** with:
 - Student name, course title, hours completed, instructor name, completion date, unique ID (`PHARM-YYYY-XXXXXXX`).
 - Landscape printable template (company green branding).
 - **View & print certificate** from the completion area and celebration modal.
 - Route: `/student/course/[courseId]/certificate` (browser Print → PDF).

3.3 Engagement

- **Meetings** — Book tutors or mentors; Jitsi join when scheduled.
- **Messages** — Chat threads with tutors/admins (when threads exist).
- **Notifications** — In-app bell for assignments, meetings, and events.
- **Profile** — Editable profile including country (used for currency display).
- **Streak metric** — Learning streak on dashboard.

3.4 Multi-currency display & payments (new)

- **NGN** is the stored price on courses (kobo/minor units).
 - **Display currency** — Nigeria (geo or profile) → NGN; international users → **USD** (live FX conversion, cached).
 - **Paystack** always charges **NGN** for MVP; USD users see a clear note that checkout is in Naira at the current rate.
 - Geo: Vercel `x-vercel-ip-country` ; profile `country` overrides when set.
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4. Tutor features (implemented)

4.1 Course studio

- Multi-step **new course** wizard (draft creation).
- **Curriculum editor v2** — Sections, drag-and-drop order, lessons (video/article), section quizzes, assignments, resources.
- **Pricing** — NGN prices in kobo; free or paid courses.
- **Landing / basics / promo video** — Course marketing fields and uploads (R2).
- **Submit for review** — Workflow to admin; email notifications.
- **Preview** — Student-style catalog preview.
- **Course overview** — Announcements and forum management.
- **Settings** — Delete course (policy-guarded), metadata.

4.2 Operations

- **Assignments** — Create, grade, notify students.

- **Communication** — Messages, announcements, forums, meetings.
- **Performance** — Overview, revenue, payment sales list, reviews, students list.
- **Payouts** — Paystack bank account resolve, wallet balance, withdrawal requests.

4.3 Partial / planned studio pages

Some manage screens are informational or stubbed (promotions/coupons, accessibility checkboxes without save, captions without schema, film URL client-only). Core authoring path (curriculum + pricing + submit) is fully functional.

5. Mentor features (implemented)

- Dashboard with pending meeting requests and availability summary.
 - **Meetings** — Accept/reject/complete; Jitsi join.
 - **Profile** — Submit for admin approval before students can book (`mentorProfileStatus` , `isActive`).
 - No course or payout surfaces (by design).
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6. Admin features (implemented)

- **Dashboard** — KPIs, enrollment trends, course status chart, sales, workflow feed.
 - **Course approvals** — Approve → published; reject with reason; emails to tutor.
 - **People CRM** — Students, tutors, mentors: search, activate/deactivate, direct message, delete (with audit log writes).
 - **Mentor applications** — Review queue for mentor profiles.
 - **Payments** — Transaction ledger, withdrawal approve/reject (Paystack transfer), platform fee % and min withdrawal settings.
 - **Messages** — Inbox and broadcast notifications.
 - **Badges** — Catalog management and rules reference.
 - **Course preview** — Admin read-only catalog view.
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7. Technical architecture

```
<pre class="code-block">Browser (Student / Tutor / Mentor / Admin)
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Next.js App Router (React Server Components + API routes)
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├─ NextAuth (JWT sessions, credentials + optional Google/Apple OAuth)
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├─ Prisma ORM → PostgreSQL (Neon)
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├─ Paystack API (charge, verify, webhook, transfers)
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├─ Cloudflare R2 (lesson media, avatars, uploads)
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├─ Resend (transactional email)
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└─ Hugging Face (AI quiz & course chat, optional)</pre>
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Security patterns

- Role enforced per API route (`auth()` + `UserRole` checks).
- Page routes guarded by `proxy.ts` for portal prefixes (`/student` , `/tutor` , `/mentor` , `/admin`).
- Paystack webhook HMAC verification; server-side transaction re-verify on purchase completion.
- Paid course access gated by `CoursePurchase SUCCESS` before content, assignments, forum, and AI features.

Key data models

User, Course, CourseSection, Lesson, Enrollment, LessonProgress, CoursePurchase, CourseCertificate, CourseReview, Assignment, AssignmentSubmission, Badge, StudentBadge, MeetingRequest, Meeting, ChatThread, PlatformSettings, WithdrawalRequest, TutorPayoutAccount, CourseApprovalWorkflow, and related enums.

8. Payments & commerce flow

1. Student clicks **Buy now** on a priced course.

2. `POST /api/payments/paystack/initialize` creates a pending `CoursePurchase` with charge amount (NGN kobo) and display metadata.
3. Paystack inline popup collects payment.
4. Client calls `POST /api/payments/paystack/verify`; webhook `charge.success` also completes purchase.
5. Enrollment created or linked; student gains content access.
6. Tutor wallet credited net amount after platform fee on success.

Tutor withdrawals: request → admin approve → Paystack transfer.

9. Certificate issuance flow

1. Student completes all required lessons and clicks **Complete course**.
2. `Enrollment.status` → `COMPLETED`; `CourseCertificate` row created with stable certificate number.
3. Student sees **View & print certificate** under completion CTA and in celebration modal.
4. Certificate page renders Pharm LMS template; **Print certificate** uses browser print (landscape PDF).

Backfill script for legacy completions: `pnpm dlx tsx scripts/backfill-course-certificates.ts --apply`

10. Deployment & configuration

Variable	Purpose
<code>DATABASE_URL</code>	PostgreSQL
<code>AUTH_SECRET</code>	NextAuth
<code>NEXT_PUBLIC_PAYSTACK_PUBLIC_KEY</code> / <code>PAYSTACK_SECRET_KEY</code>	Payments
<code>R2_*</code>	Media storage
<code>RESEND_API_KEY</code> / <code>EMAIL_FROM</code>	Email
<code>DEFAULT_DISPLAY_CURRENCY</code> / <code>FX_CACHE_TTL_SECONDS</code> / <code>EXCHANGERATE_API_KEY</code>	Multi-currency
<code>HUGGINGFACE_API_KEY</code>	AI features (optional)

Recommended host: **Vercel** (geo headers for currency). Package manager: **pnpm**.

11. Quality status & known gaps

Strengths

- End-to-end learning, payments, approvals, and tutor payouts.
- Rich curriculum and assessment tooling.
- Certificates and multi-currency improve international readiness.

Known gaps (prioritized)

Priority	Item
High	Enforce <code>isActive</code> at login/API; pending checkout idempotency; FX API fallback
Medium	Student purchase history page; real catalog ratings (remove placeholder stars); notifications page
Medium	Certificate public verify URL; admin txn display currency fields
Low	Tutor studio stub pages (promotions, accessibility, captions); engagement analytics placeholders
Low	Automated test suite

12. Roadmap (suggested)

- **Trust & money** — Security hardening, payment edge cases, webhook error handling.
 - **Learner polish** — Purchase receipts, assignments in player, leaderboard filters.
 - **Credentials** — Public certificate verification page, optional PDF server generation.
 - **Enterprise** — SSO, analytics dashboards, audit log UI, formal exam/question banks.
 - **International** — Paystack USD settlement when merchant account supports it.
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13. Conclusion

Pharm LMS delivers a complete pharmacy-training platform: governed course publishing, paid and free enrollment, structured learning with AI aids, mentor meetings, gamification, **Paystack commerce**, **multi-currency pricing display**, and **printable completion certificates** under the Pharm LMS brand. The codebase is structured for continued delivery on security, analytics, and accreditation-focused features.

Pharm LMS © 2026 — Company-owned learning platform.